Observer Patch Holography Cosmological Vacuum and Structure Formation: Finite Transfer Conditions, Fluctuation Ensembles, and Proto-Objects

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Abstract

A finite quotient, a reference measure and an action determine an observer ensemble. A vacuum interpretation needs additional reversible Euclidean data and compatible refinement. We give the corresponding finite transfer construction and separate it from stationary repair and from supplied Hamiltonian ground states. Native density records have a reflection-positive Gaussian reconstruction generated by the screen operator A, while the full return reference approaches the normalized trace. An active free field has vacuum covariance $(2\sqrt{B})^{-1}$ under its declared preparation; its flat massless spectrum grows as k^2 per logarithmic interval and therefore is not a scale-invariant primordial curvature spectrum. Structure formation requires an expanding background, a curvature readout, and transfer dynamics on the same state. The capacity relation $\Lambda\ell_P^2 = 3\pi/N$ is a conditional horizon-entropy dictionary, not an independent selection of the cosmological constant.

1 Ensemble, vacuum and background

A probability law on a finite physical quotient is distinct from a Hamiltonian vacuum and from cosmological initial data. The native source screen has a positive continuum operator of spectral dimension two; its construction supplies no expanding background or physical screen-to-volume map [6, 4, 7].

If the capacity N is identified with de Sitter horizon entropy in nats, then

$$\Lambda\ell_P^2 = \frac{3\pi}{N}.$$

This is the entropy relation read backwards; it determines Λ only after N and the area interpretation are supplied. For a separately conserved vacuum component with $\rho \propto N^{-1}$, the continuity equation gives $w = -1 + \frac{1}{3} d\log N/d\log a$. Constant N gives $w = -1$; nondecreasing N gives $w \geq -1$. Neither relation chooses N. This w is the effective-fluid quantity of the continuity equation: consensus about identical fixed-volume records cannot select an unobserved strain derivative, since $H_\alpha(V) = Q(V/V_0)^{-\alpha}$ gives $p/\rho = \alpha$. Three geometrically spaced energy readings identify the response within a family with $\alpha \geq 0$, which excludes $w = -1$ at positive energy, and an additive energy offset changes

p/ρ while leaving pressure unchanged [3]. Native regional entropy grows with port count, whereas mutual information per crossing seam vanishes, so the whole-screen horizon reading needs a region map [11]. Target-informed capacity formulas and their retrospective data comparisons belong to the likelihood companion [1]. A smooth Einstein interpretation also needs the common-stress and tensor-curvature or small-ball identification premises [2].

Radiation-era capacity histories. With conserved matter and ideal radiation, the gamma-one future-event branch obeys $d\Omega_x/d\log a = \Omega_x(1-\Omega_x)(1+r+2\sqrt{\Omega_x})$, where $r = \rho_r/(\rho_m+\rho_r)$. An outward Taylor enclosure controls the branch through $z = 10^6$. Its early radiation-era fraction scales as a^2, whereas fixed capacity gives a^4 and the Hubble-area control retains a constant fraction. The present amplitudes and radiation content remain supplied; these are conditional expansion histories, not derived particle abundances. The defining source and outward enclosures are specified in `model/openwave_radiation_capacity`.

Perturbations of a declared capacity background. Put $x = \sqrt{\Omega_x}$ and $r = \rho_r/(\rho_m+\rho_r)$. The conserved event-area branch has

$$c_a^2 = \frac{-2-3x+3x^2+2x^3+rx(1+x)}{6}.$$

The supplied past-background enclosures imply $c_a^2 < -1/5$, rejecting an adiabatic barotropic-fluid attachment. This instability is consistent with the holographic-fluid analysis of Myung. The established holographic-quintessence reconstruction of Zhang instead supplies a canonical scalar with

$$\frac{d\phi}{d\log a} = M_{\mathrm{Pl}}\sqrt{2\Omega_x(1-x)}, \qquad V = \frac{\rho_x(2+x)}{3}, \qquad c_s^2 = 1.$$

For this radiation-inclusive past domain, exact positive Bernstein coefficients and the Lean certificate give

$$2 \le \frac{V''}{H^2} \le 6.$$

This is a stable local scalar completion of the declared background, not a source-derived particle mass or a cosmological likelihood fit. The supplied amplitudes and horizon choice remain, and native-source transfer and perturbation-data acceptance are unproved. The polynomial bounds are in `Lean/ObserverPatchHolography/CapacityScalarCurvature.lean`; the complete background and controls are in `model/openwave_capacity_perturbations`.

Constrained scalar, matter and radiation transfer. The canonical capacity scalar has a declared linear attachment to Einstein gravity, ideal dust and ideal radiation. Put $N = \log a$ and $h = H_N/H$. In conformal Newtonian gauge, with zero anisotropic stress, write $ds^2 = a^2[-(1+2\Phi)d\tau^2 + (1-2\Phi)d\mathbf{x}^2]$ and use the reduced Planck mass M_{Pl}. The perturbation conventions are those of Ma and Bertschinger. Set $m = \Omega_m$, $r = \Omega_r$, $s = \phi_N/M_{\mathrm{Pl}}$, $A = V_\phi/(H^2 M_{\mathrm{Pl}})$, $\mathcal{M} = V_{\phi\phi}/H^2$, $Q = k^2/(aH)^2$, $U_i = aH\theta_i/k^2$, $y = \delta\phi/M_{\mathrm{Pl}}$ and $z = y_N$. The background has $h = -(3m+4r+s^2)/2$ and $s_N + (3+h)s + A = 0$. The full linear equations are

$$\begin{aligned}
P := \Phi_N &= -\Phi + \tfrac{1}{2}(3mU_m + 4rU_r + sy), \\
\delta_{m,N} &= -QU_m + 3P, & U_{m,N} &= hU_m + \Phi, \\
\delta_{r,N} &= -\tfrac{4}{3}QU_r + 4P, & U_{r,N} &= (1+h)U_r + \delta_r/4 + \Phi, \\
y_N &= z, & z_N &= -(3+h)z - (Q+\mathcal{M})y + 4sP - 2A\Phi.
\end{aligned}$$

The first equation is the momentum constraint. With

$$C = Q\Phi + 3(P + \Phi) + \tfrac{1}{2}(3m\delta_m + 3r\delta_r + sz - s^2\Phi + Ay),$$

exact substitution gives both $C_N = -(3 + 2h)C$ and

$$\Phi_{NN} + (4 + h)\Phi_N + (3 + 2h)\Phi = \tfrac{1}{2}(r\delta_r + sz - s^2\Phi - Ay).$$

Consequently $C(a) = C(a_0)(a_0/a)^3 H(a_0)^2/H(a)^2$; numerical constraint projection is unnecessary. The regular zero-wave-number adiabatic mode uses $T = (3/2)(H/a)\int_0^a da'/H(a')$, $\Phi = 3/2 - T$, $U_m = U_r = T$, $y = sT$ and $z = s_N T + s(\Phi + hT)$, normalized to unit asymptotic radiation-era potential. At finite k, the supplied initial correction $\delta_r = -4T - 2Q\Phi/(3r)$, with $\delta_m = -3T$, enforces $C = 0$. Integrating $\Delta_m = \delta_m + 3U_m$, $\Delta_r = \delta_r + 4U_r$ and $U_r - U_m$ avoids subtraction of order-one superhorizon variables. This normalization defines a linear transfer coefficient, not a physical unit-amplitude fluctuation or a primordial-spectrum fit.

Regular growing initial data, an earlier-start comparison, tolerance refinement and an independent Einstein–de Sitter solution check the numerical transfer. With the inherited background inputs, present comoving matter transfer relative to a same-density cosmological-constant control is

k/H_0	0.1	1	10	100
$T_m/T_{m,\Lambda}$	0.97585	0.97001	0.91494	0.90822

These conditional differences supply no observed match or cosmic microwave background fit. Native scalar and primordial-state derivations are absent; free-streaming species and recombination are omitted. The constraint identities are exact, while the numerical solutions have convergence diagnostics without interval error certificates (`model/openwave_capacity_transfer`).

2 Matter and States on a Supplied Spatial Domain

The Whitney construction on the fixed tetrahedral cone gives a controlled matter continuum in a specified sector of one charged scalar action. For $m^2 > 0$, $g \geq 0$, a supplied $u \in C^2([0,T]; H^2(\Omega))$ solving the real cubic Klein–Gordon equation with natural Neumann boundary, and Ritz-projected initial data, uniform conforming refinements give

$$\sup_{0\leq t\leq T} (\|u_n - u\|_{H^1} + \|\dot{u}_n - u_t\|_{L^2}) \leq C_T n^{-1}.$$

The real, zero-current solutions are invariant solutions of the full charged action. The proof uses a fixed shape-regular refinement family, the conserved nonlinear energy and a Ritz error estimate, as proved in *Observers Are All You Need* [3].

On the original finite mesh with nonzero coupling e, eliminating mean-zero gauge directions gives a complete kinetic metric γ on $\mathcal{Q} \cong \mathbb{R}^{30} \times \mathbb{C}^{13}$. With supplied $\hbar > 0$ and Laplace–Beltrami ordering, the neutral subspace of $L^2(\mathcal{Q}, d\mathrm{vol}_\gamma)$ has a unique nonnegative self-adjoint Hamiltonian and unitary dynamics. Normalized Gaussian half-density states lie in its operator domain and have explicit initial matter moments [3]. These are states of a finite spatial discretization, not a derived cosmological ground state or a continuum quantum field limit.

For the same autonomous action, the Jacobi duration

$$d\tau = \sqrt{\frac{G_q(dq, dq)}{2(E - V(q))}}$$

recovers action time on regular nonturning solutions at supplied energy E, using the full coupled kinetic metric or its gauge reduction [3]. Its configuration-based reconstruction supplies an internal model clock. A cosmological interpretation additionally requires observer-selected geometry, physical preparation and units, and a compatible expanding background. Those attachments are separate from these matter and state theorems.

The golden source population of the spacetime construction carries a second matter and state construction on the same population that the causal order uses. Its positive source-metric scalar action has a finite-time continuum limit with an explicit detector error bound, a canonical quantization whose quadratic Hamiltonian has a unique normalized ground state, the original action vacuum, and a bounded probability comparison between the split evolution of that vacuum with a coherent displacement and the continuous evolution of the same finite action, with all sixty-four modes retained at $q = 5$. The joint spatial and temporal refinement of one compact detector is certified at $q = 233$, with total error below 0.0501 against a resolved response above 0.1377. This is a vacuum state of a finite scalar action on the source population, with the population, action and tick supplied. It is neither a reflection-positive Euclidean vacuum of a cosmological release law nor a source-instantiated triple $(J_r, V_r, a_{t,r})$ in the sense of the next section, and it carries no expanding background.

On the same family a spatially flat expanding background is a record density: the scale factor enters only as the weight a^4 per comoving cell and conformal tick, and the redshift is the fourth root of a record-count ratio of congruent comoving windows with constant scale factor within each, or in the shrinking-window limit of a continuous profile. The finite-source cosmology paper proves these statements with their Lean module [5]. The record-production profile is a supplied datum; the vacuum and fluctuation conditions below concern the field on that background and do not select the profile.

3 Finite Quotient Ensemble Theorem

Let $Q_r = \Sigma_r/\Gamma_r$ be a finite physical quotient and $n_r : Q_r \to N_r$ a retraction onto settled states. Then $(n_r)^2_{\#} = (n_r)_{\#}$, and every law supported on N_r is fixed. Settlement alone chooses no ensemble. A declared positive base weight m_r and action S_r, finite somewhere, instead give

$$\mu_r(q) = \frac{m_r(q)e^{-S_r(q)}}{Z_r}, \qquad Z_r = \sum_q m_r(q)e^{-S_r(q)} > 0.$$

A prior pushed through n_r agrees with this law only when its normal-form fibre sums equal μ_r. Uniform sampling of representatives weights quotient classes by their multiplicities; a conditional lift must instead use $\widetilde{\mu}(x) = \mu(\pi x)\kappa(x \mid \pi x)$, with unit conditional mass on each fibre.

Theorem 3.1 (Finite selection and refinement)**.** *For quotient observables F_a with a feasible full-support expectation constraint, relative entropy maximization yields $\mu(q) \propto m(q)e^{-\sum_a \theta_a F_a(q)}$. For a faithful quantum reference σ, the analogous state is $\rho \propto \exp(\log\sigma - \sum_a \theta_a F_a)$. On a proper joint support P, compress $\log\sigma$ before exponentiating; $\log(P\sigma P)$ cannot replace that compression without a commutation hypothesis.*

For compatible coarse maps $c_{sr} : Q_s \to Q_r$, exact refinement consistency is equivalent to

$$\sum_{c_{sr}(q')=q} m_s(q')e^{-S_s(q')} = \alpha_{sr} m_r(q)e^{-S_r(q)}, \qquad \alpha_{sr} > 0.$$

One-step total-variation defects $\delta_{k+1,k}$ accumulate by at most $\sum_{k=r}^{n-1}\delta_{k+1,k}$. For a bounded observable O, sampling error ϵ_{samp} and total refinement error ϵ_{ref} give expectation error at most $2\|O\|_\infty(\epsilon_{\mathrm{samp}}+\epsilon_{\mathrm{ref}})$.

Proof. The multiplier equations and strict concavity give the entropy maximizers. Normalizing the fibre sums proves the equivalence; contractivity of pushforward in total variation and the triangle inequality give the accumulated bound. The expectation estimate is the dual bound for total variation. □

A tracially pointed gauge quotient additionally fixes representation multiplicities: if the chosen sector is $\bigoplus_\alpha V_\alpha\otimes M_\alpha$, its central weights are $d_\alpha m_\alpha/\sum_\beta d_\beta m_\beta$. These weights are invariant under trace-preserving intertwiners and inert trivial ancillas, not arbitrary changes of representation multiplicity. A representative Markov kernel descends only if its transition sums to each quotient fibre are independent of the input representative. An asymmetric proposal R uses acceptance $\min\{1, w(q')R(q',q)/[w(q)R(q,q')]\}$, with $w = me^{-S}$, to sample the specified law. If a coarse conditional free energy differs from the claimed exponential family by at most ε uniformly, normalization bounds its total-variation defect by $\tanh\varepsilon$.

For $L_{\mathrm{rep}} = \sum_C c_C(I-E_C)$, let $\mathcal{F} = \bigcap_C \operatorname{Ran} E_C$. If $c_C \geq \gamma_* > 0$,

$$L_{\mathrm{rep}} \geq \gamma_*\kappa(I-P_{\mathcal{F}}), \qquad \kappa = \inf_{0\neq f\perp\mathcal{F}} \frac{\sum_C \|(I-E_C)f\|^2}{\|f\|^2} > 0$$

when $\mathcal{F}^\perp \neq 0$. Finite-dimensional compactness proves positivity. Replacing $\mathcal{F}$ by constants needs ergodicity; a bound uniform in refinement is separate. The expectations contain a specified reference law, and multiplying all rates changes the clock and gap without selecting that law.

Theorem 3.2 (Finite Euclidean vacuum)**.** *Supply $(Q_r, m_r^0, J_r, V_r, a_{t,r})$, with $m_r^0 > 0$, a connected graph of symmetric nonnegative conductances J_r, a real potential V_r and $a_{t,r} > 0$. On $L^2(Q_r, m_r^0)$,*

$$(H_r^E f)(q) = \frac{1}{m_r^0(q)}\sum_{q'} J_r(q,q')[f(q)-f(q')] + V_r(q)f(q)$$

has a unique positive normalized ground state Ω_r. Its law is $\mu_r^{\mathrm{vac}} = |\Omega_r|^2 m_r^0$; conjugating $e^{-a_{t,r}(H_r^E - E_{0,r})}$ by Ω_r gives a stochastic kernel reversible for this law.

Proof. The symmetric conductance form is nonnegative and the finite potential is bounded below. Connectedness makes the finite Feynman–Kac semigroup positivity improving. Perron–Frobenius gives the ground state; its eigenvector equation proves the row-sum identity for the conjugated kernel. Self-adjointness proves detailed balance. □

The Euclidean data must be selected independently of the desired vacuum law. Otherwise a positive transfer operator can be constructed around any faithful target distribution. A continuum vacuum further requires reflection positivity or an equivalent reconstruction and compatible transfer refinement.

Native source screens. On the native source screens the actual density law has the reflection-positive reconstruction that a continuum interpretation requires, along one common refining subsequence; it supplies no coherent preparation, relativistic vacuum or Lorentz symmetry [8] (Section 4). The same continuum energy is not boundedly congruent to the static boundary action of a smooth compact three-dimensional Dirichlet bulk, whose Steklov eigenvalues grow like $\sqrt{n}$ while the native eigenvalues grow like n, and internal rank-three carrier modes vanish from bounded-energy families; a radial lift needs a radial observation algebra that the native energy does not supply [7].

4 Vacuum Claim Boundary

A simulated field is classified as seed noise, repair jitter, or a conventional reference ensemble unless it satisfies the finite quotient ensemble conditions. An OPH-native vacuum requires source data of the form

$$\mathfrak{S}_r^E = (Q_r, m_r^0, J_r, V_r, a_{t,r})$$

with J_r, V_r, and $a_{t,r}$ derived from OPH source structure rather than from a target law or a desired visual spectrum. Directed accepted repair moves cannot be reused as reversible Euclidean conductances. The native seam rule differs: each unread repair has the law of a fair identity-or-endpoint swap, so its configuration transfer K is symmetric and positive, and the actual native density law has a reflection-positive Gaussian reconstruction. The obstruction on the native source lies elsewhere: the reconstructed one-particle generator is the continuum operator A, not $\sqrt{A}$, and the full configuration return reference built from K approaches the normalized trace at fixed modular parameter and gives the density-wave sector vanishing weight, so it identifies no quantum vacuum [8].

4.1 Vacuum and initial-state readouts on the effective product

The active product construction has a different, conditional state contract. With its positive spatial operator B, canonical fields and supplied inverse temperature b, its ordered free-field two-point function is

$$C_b(t) = \frac{\coth(b\sqrt{B}/2)\cos(t\sqrt{B}) - i\sin(t\sqrt{B})}{2\sqrt{B}}.$$

The limit $b \to \infty$ gives the static ground-state covariance $(2\sqrt{B})^{-1}$; finite b changes fluctuations while preserving the same commutator and response. These are joint observer-readout predictions of the declared free-field family, not the passive native density law or the interacting field's covariance [8]. The radial apparatus, state preparation and physical clock map remain inputs.

On a finite algebra with two invariant sectors, restricted thermal equilibrium leaves a central sector weight:

$$\rho_w = w\frac{e^{-bH_+}}{Z_+} \oplus (1-w)\frac{e^{-bH_-}}{Z_-}, \qquad 0 < w < 1.$$

Every such state has the same restricted modular flow. Cooling each block does not select w. The full Gibbs preparation additionally chooses $w = Z_+/(Z_+ + Z_-)$; its global ground-state limit is therefore a stronger state choice. Admitted record statistics can test sector populations, but do not identify an unobserved purification.

A concrete interacting preparation is available on the 90 localized CAR modes of the common defect/current construction. Its Hamiltonian $H = H_{\text{pair}} + \lambda(N_W - 45)^2$, $\lambda \geq 0$, preserves parity and has a faithful finite Gibbs state $\rho_\beta = e^{-\beta H}/Z$ for $\beta > 0$. A global spectral reset prepares this state. For $H_s = H + sX$ at fixed β, let $E_s = \operatorname{Tr}(\rho_s H_s)$ and $S_s = -\operatorname{Tr}(\rho_s \log \rho_s)$. Derivatives at $s = 0$ obey

$$\rho'_\beta = -\beta \int_0^1 \rho_\beta^t (X - \langle X \rangle)\rho_\beta^{1-t}\, dt, \qquad S' = \beta(E' - \langle X \rangle).$$

This is a chosen preparation on the same interacting Hamiltonian, not a cosmological vacuum or a geometric stress variation. It does not select sector weights from native repair, an expanding background or a physical clock [9].

A direct cosmological identification has a discriminating control. In the flat massless limit, with $\hbar = c = 1$ and $\langle \phi_{\mathbf{k}} \phi_{\mathbf{k}'} \rangle = (2\pi)^3 \delta(\mathbf{k} + \mathbf{k}') P_\phi(k)$, the static vacuum gives

$$P_\phi(k) = \frac{1}{2k}, \qquad \Delta^2_\phi(k) = \frac{k^3 P_\phi(k)}{2\pi^2} = \frac{k^2}{4\pi^2}.$$

Thus a scale-independent identification of this field with primordial curvature does not reproduce the nearly scale-invariant curvature spectrum inferred from CMB data [10]. This flat-limit control is not a Fourier-spectrum claim for the bounded native radial product. An expanding evolution and an operational curvature map must supply the transfer before the state can predict cosmological observations. The initial-state target is consequently a law for accessible earlier-state correlations and their later records, not an assumed observable absolute beginning of the Universe.

5 From a state to structure formation

The state constructions above do not fix a primordial curvature field. A complete cosmological model must supply the source Euclidean data or an equivalent physical preparation, the expanding geometry and clock, and the map from its accessible field to conserved stress and curvature. A screen covariance alone leaves an infinite-dimensional radial null space; physical source dilation or complete radial cross-covariances can provide uniqueness under their additional hypotheses [5]. The resulting initial data must then enter one compatible matter, photon and metric transfer system and a stated observational likelihood. Quotient invariance, stationarity of a sampler or visually coherent trajectories establishes none of these physical identifications by itself.

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